USER MANAGEMENT SYSTEM

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FULL-STACK WEB APPLICATION DOCUMENTATION

1. Application Description:

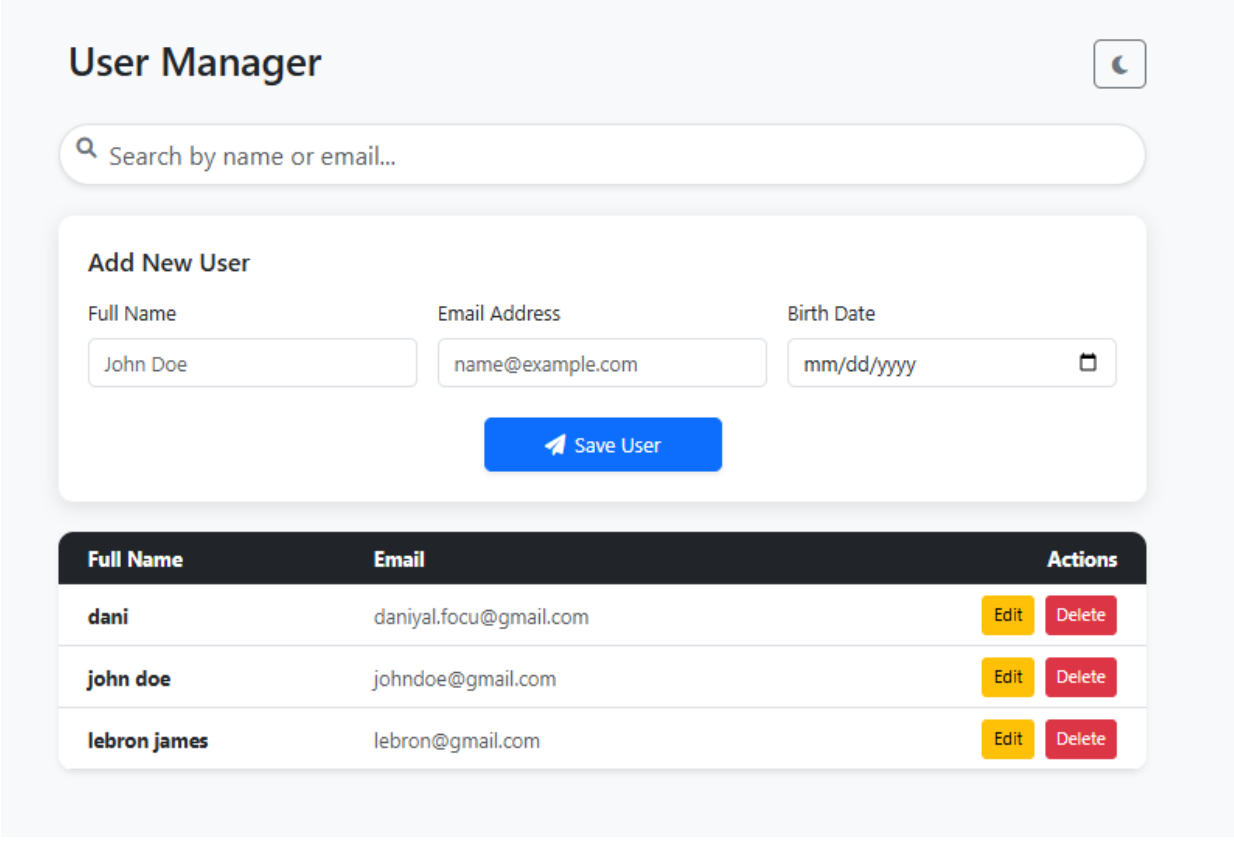
This application is a robust, full-stack solution designed to manage user records efficiently. It features a modern **ASP.NET Core 8 Web API** backend paired with a responsive **JavaScript/Bootstrap 5** frontend. The system implements a full suite of RESTful services to ensure seamless data management and persistence.

2. Features Implemented:

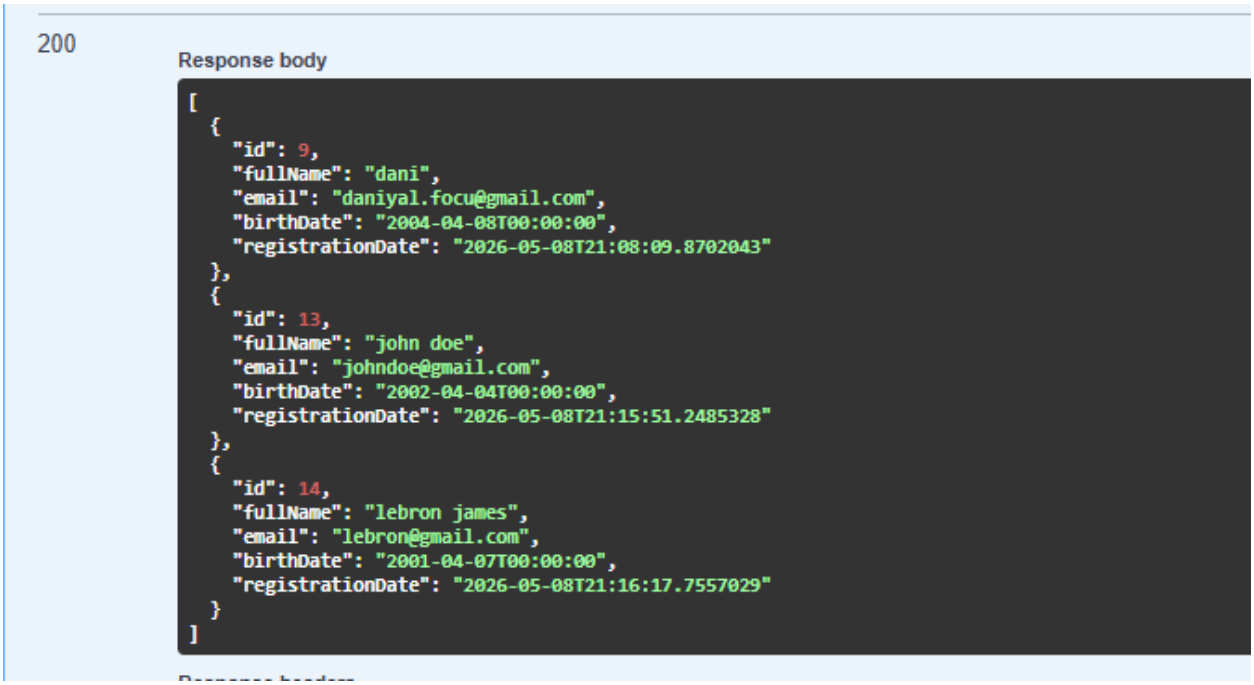
- **Complete CRUD Operations:** Users can be listed, added, edited, and deleted through a unified interface.
- **Interactive UI:** Includes a Dark/Light mode toggle with dynamic icons and SweetAlert2 notifications.
- **Real-time Search:** A client-side filtering system for instant user lookup by name or email.
- **Data Integrity:** Backend validation blocks duplicate emails, while frontend logic prevents future birth dates.
- **Persitence Storage:** All data is securely stored in a local SQLite database.

3. Visual Interface:

Main Dashboard:



API Testing Via Swagger:



4. Technical Implementation:

RESTful API Architecture

Method	Endpoint	Description
GET	/api/users	Retrieves the list of all users.
GET	/api/users	Retrieves a specific user by ID.
POST	/api/users	Creates a new user record.
PUT	/api/users	Updates an existing user record.
DELETE	/api/users	Removes a user from the database.

5. Instructions for Running:

To run the application locally, follow these steps:

1. Navigate to the project root directory **`UserManagementApp`** .
2. Open a terminal and run the command: **`dotnet run`** .
3. Open a web browser and navigate to : **`http://localhost:5074`** (or the port specified in your terminal).

The database file (**`users.db`**) will be automatically generated upon the first execution.

6. Improvements & Known Issues:

Known Issues: None currently identified in the core CRUD workflow.

Suggested Improvements:

- Server-Side Validation
- Expanding the database schema to include user profile pictures.
- Export to CSV/Excel